# Chapter

# Matrices

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1. If $A = [-1 \ \ 2]$ and $B = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$

Which of the following operation is possible?

(a) $A$ - $B$ (b) $A + B$

(c) $AB$ (d) $BA$

SQP 2024

2. If matrix $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, then the value of $x$ is:

(a) 2 (b) 4

(c) 8 (d) 10

Main 2024

3. If $= \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, the value of $x$ and $y$ respectively are :

(a) 1, -2 (b) -2, 1

(c) 1, 2 (d) -2, -1

Main 2023

4. If $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix}$, and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Find $A(B + C) - 14I$.

Main 2023

5. Find the values of $x$, $y$, $a$ and $b$ if

$$\begin{bmatrix} x+y & a+b \\ a-b & x-3y \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 3 & -7 \end{bmatrix}$$

Comp 2023

6. If $A = \begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{-1} & \mathbf{7} \end{bmatrix}$ then $A^2$ is :

(a) $\begin{bmatrix} 4 & 0 \\ 1 & 49 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 0 \\ -9 & 49 \end{bmatrix}$

(c) $\begin{bmatrix} 4 & 0 \\ 9 & 49 \end{bmatrix}$

(d) $\begin{bmatrix} 4 & 9 \\ -9 & 49 \end{bmatrix}$

Main 2022 Sem I

7. If $A = \begin{bmatrix} \mathbf{3} & \mathbf{5} \\ \mathbf{1} & \mathbf{4} \end{bmatrix}$, $B = \begin{bmatrix} \mathbf{2} & \mathbf{4} \\ \mathbf{0} & \mathbf{3} \end{bmatrix}$, $C = \begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{2} & \mathbf{1} \end{bmatrix}$ then $5A$ - $BC$ is :

(a) $\begin{bmatrix} -5 & -23 \\ 1 & 17 \end{bmatrix}$

(b) $\begin{bmatrix} 5 & 23 \\ 1 & 17 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 8 \\ -3 & 3 \end{bmatrix}$

(d) $\begin{bmatrix} 5 & 23 \\ -1 & 17 \end{bmatrix}$

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**Solution**

**1.**

**Ans: (c)** $AB$

**Explanation:**

Matrix $A$ is of order $1 \times 2$ and matrix $B$ is of order $2 \times 2$.

- Addition or subtraction of matrices is possible only when the matrices are of the **same order**.

  Hence, $A + B$ and $A$ - $B$ are **not possible**.

- Matrix multiplication $A$ - $B$ is possible if the **number of columns of $A$** is equal to the **number of rows of $B$.**

Here,

$$A : 1 \times 2 \text{ and } B : 2 \times 2$$

Since the number of columns of $A$ equals the number of rows of $B$,

$$AB \text{ is possible.}$$

However, $BA$ is not possible since the number of columns of $B$ is not equal to the number of rows of $A$.

**Thus, option (c) is the correct answer.**

**2.**

**Ans: (c)** 8

**Explanation:**

We have

$$A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

$$A^2 = A \times A$$

$$= \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2(2)+2(0) & 2(2)+2(2) \\ 0(2)+2(0) & 0(2)+2(2) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

Comparing with

$$A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix},$$

we get

$$x = 8$$

**Thus, option (c) is the correct answer.**

**3.**

**Ans: (a)** 1, −2

**Explanation:**

By definition of equality of matrices, the corresponding elements of equal matrices are equal.

$$2x = \text{and } 4y = -8$$

$$x = 1 \text{ and } y = -2$$

**Thus, option (a) is the correct answer.**

**4.**

**Ans:**

First find $B + C$,

$$B + C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 9 \end{bmatrix}$$

Now,

$$A(B+C)=\begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}\begin{bmatrix}5 & 3\\3 & 9\end{bmatrix}$$

$$=\begin{bmatrix}1(5)+3(3) & 1(3)+3(9)\\2(5)+4(3) & 2(3)+4(9)\end{bmatrix}$$

$$=\begin{bmatrix}14 & 30\\22 & 42\end{bmatrix}$$

Now,

$$14I=14\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}=\begin{bmatrix}14 & 0\\0 & 14\end{bmatrix}$$

Therefore,

$$A(B+C)-14I=\begin{bmatrix}14 & 30\\22 & 42\end{bmatrix}-\begin{bmatrix}14 & 0\\0 & 14\end{bmatrix}$$

$$=\begin{bmatrix}0 & 30\\22 & 28\end{bmatrix}$$

$$\boxed{A(B+C)-14I=\begin{bmatrix}0 & 30\\22 & 28\end{bmatrix}}$$

**5.**

**Ans:**

By the definition of equality of matrices, the corresponding elements of equal matrices are equal.

$$x+y=5 \qquad (1)$$

$$a+b=-1 \qquad (2)$$

$$a-b=3 \qquad (3)$$

$$x-3y=-7 \qquad (4)$$

Adding equations (2) and (3),

$$2a = 2 \Rightarrow a = 1$$

Substituting $a = 1$ in (2),

$$1 + b = -1 \Rightarrow b = -2$$

Now subtracting equation (4) from (1),

$$(x + y) - (x - 3y) = 5 - (-7)$$

$$4y = 12 \Rightarrow y = 3$$

Substituting $y = 3$ in (1),

$$x + 3 = 5 \Rightarrow x = 2$$

**Hence,**

$$\boxed{x = 2,\ y = 3,\ a = 1,\ b = -2}$$

**6.**

**Ans: (b)** $\begin{bmatrix} 4 & 0 \\ -9 & 49 \end{bmatrix}$

**Explanation:**

We have

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix}$$

$$A^2 = A \times A$$

$$= \begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2(2) + 0(-1) & 2(0) + 0(7) \\ -1(2) + 7(-1) & -1(0) + 7(7) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 \\ -9 & 49 \end{bmatrix}$$

**Thus, option (b) is the correct answer.**

7.

Ans: (d) $\begin{bmatrix} 5 & 23 \\ -1 & 17 \end{bmatrix}$

Explanation:

$$5A = 5\begin{bmatrix} 3 & 5 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 15 & 25 \\ 5 & 20 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2(1)+4(2) & 2(-1)+4(1) \\ 0(1)+3(2) & 0(-1)+3(1) \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 6 & 3 \end{bmatrix}$$

$$5A - BC = \begin{bmatrix} 15 & 25 \\ 5 & 20 \end{bmatrix} - \begin{bmatrix} 10 & 2 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 23 \\ -1 & 17 \end{bmatrix}$$

Thus, option (d) is the correct answer.